College Board AP® Chemistry

Your notes

Solutions & Mixtures

Contents

- Calculations About Solutions
- Homogeneous & Heterogeneous Mixtures
- Molarity
- Particulate Representations of Solutions
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Calculations About Solutions

- Usually in calculations involving solutions, we are required to determine:
  - Number of moles or mass of solute
  - Number of moles of ions that make up the solute
  - Molarity of solution
  - Volume of solution
- The calculation would involve dilution and require the use of the dilution formula
  - Dilution formula: $M_s \times V_s = M_d \times V_d$
  - This is different from calculations involving mixing two solutions containing the same solute or with a common ion and being asked to determine the concentration of the mixed solution or the common ion

Worked Example

What volume of 4.0 M lead(II) nitrate solution, $Pb(NO_3)_2$ (aq), contains 0.50 mol of Pb^{2+}? What is the number of moles of NO_3^- ions present in this solution?

Analyse:

- We are provided with the concentration of the solution (4.0 M) and the number of moles of Pb^{2+} ions (0.50 mol)
- We are required to determine the volume of the solution and the number of moles of the nitrate ions

Plan:

- First, we determine the number of moles of the $Pb(NO_3)_2$ solute from the number of moles of Pb^{2+} ions using the mole ratio from the dissociation ionic equation
- Use the same mole ratio to determine the number of moles of nitrate, NO_3^-, ions
- Then use the calculated number of moles of the solute and the concentration to determine the volume of the solution

Answer:

- **Step 1:** Write the dissociation equation for the solute:
  - $Pb(NO_3)_2\,(aq) \rightarrow Pb^{2+}\,(aq) + 2\,NO_3^-\,(aq)$
- **Step 2:** From the equation, the mole ratio of $Pb(NO_3)_2$: Pb^{2+} is **1:1** and Pb^{2+}: NO_3^- is **1:2**, then:
  - $n_{Pb(NO_3)_2} = n_{Pb^{2+}} = 0.50 \text{ mol}$
  - $n_{NO_3^-} = 2 \times n_{Pb^{2+}} = 2 \times 0.50 = 1.0 \text{ mol}$

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- **Step 3:** Determine the volume of the solution using the molarity expression
  - $n = M \times V$
  - $V = n/M$
  - $V = 0.50/4.0$
  - $V =$ ***0.125 L or 125 mL***



Examiner Tips and Tricks

- This overall question contains more than one question but is not split into parts a, b, etc
  - When you encounter questions like this, you do not have to answer the questions in the order they are asked
- For this question, you cannot answer the first actual question before the second one because the number of moles of the solute, $Pb(NO_3)_2$, is required to calculate the volume of the solution



Worked Example

What is the molar concentration of the KOH solution obtained from mixing two solutions of KOH, **A** and **B**?

- Solution **A:** 55.0 mL of 0.10 M KOH
- Solution **B**: 75.0 mL of 0.15 M KOH

Analyse: We are provided with the concentration and volume of two aqueous KOH solutions and asked to determine the concentration of the solution on mixing

Plan:

- Determine the number of moles of the solute in each solution using the molarity equation
- Determine the total number of moles of the solute by adding the number of moles from each solution
- Divide this total number of solute moles by the total volume of the solution

Answer:

- **Step 1:** Determine the number of moles of KOH in solutions **A** and **B:**
  - For solution **A**:
    - $n_{KOH} = M \times V$ **(Remember volume must be in L)**
    - $n_{KOH} = 0.1 \times 0.0550$
    - $n_{KOH} = 0.0055$ mol
  - For solution **B**:
    - $n_{KOH} = M \times V$ **(Remember volume must be in L)**
    - $n_{KOH} = 0.15 \times 0.0750$
    - $n_{KOH} = 0.01125$ mol



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- **Step 2:** Add up the number of moles of KOH from each solution:
  - $(n_{KOH})_T = (n_{KOH})_A + (n_{KOH})_B$
  - $(n_{KOH})_T = 0.0055 + 0.01125$
  - $(n_{KOH})_T = 0.01675$ mol
- **Step 3:** Determine the total volume of the mixture, in L:
  - $V_T = 75.0 + 55.0$
  - $V_T = 130.0/1000$
    - **Remember:** volume must be in L
  - $V_T = 0.130$ L
- **Step 4:** Determine the molarity of the mixture by dividing the total number of moles by the total volume:
  - $M_T = n_T/V_T$
  - $M_T = 0.01675/0.130$
  - $M_T =$ **0.129 M**

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Worked Example

How much water would be required to dilute 500.0 mL of 2.4 M $CaCl_2$ solution to make a 1.0 M solution?

Analyse: We are asked to determine the quantity of water required to be added to a stock solution (500 mL, 2.4 M) to give a 1.0 M dilute solution

Plan:

- Using the dilution formula, we determine the volume of the dilute solution
- Subtract the volume of the stock solution from the final volume to obtain the volume of water added

Answer:

- **Step 1:** Rearrange the dilution formula to determine the volume of the dilute solution (V_d):
  - $V_d = M_s \times V_s / M_d$
  - $V_d = 2.4 \times 500.0 / 1.0$
  - $V_d = 1200$ mL
- **Step 2:** Subtract the volume of the stock solution (V_s) from the volume of the dilute solution (V_d) to obtain the volume of water:
  - $V_{H2O} = V_d - V_s$
  - $V_{H2O} = 1200 - 500$
  - $V_{H2O} =$ **700 mLs**

Examiner Tips and Tricks

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- When using the dilution formula, ensure the volume of the dilute and stock solutions are in the **same units**
- When using the molarity expression, **always** change your volume to **liters**, **L**



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Homogeneous & Heterogeneous Mixtures

- A mixture is a material that can be separated by physical means into two or more substances
  - For example, when you dissolve sodium chloride in water, you obtain a mixture whose composition depends on the amount of sodium chloride dissolved
  - This mixture can be separated into its components by the physical process of distillation
- Mixtures are classified into two types:
  - **Heterogeneous mixtures** which consist of physically distinct components with different properties
    - For example, a mixture of potassium dichromate and iron filings which can be separated by the use of a magnet
  - **Homogeneous mixtures**, also known as **solution** consist of components with uniform properties
    - Air is an example of a homogeneous mixture of 78 % nitrogen and 21% oxygen gas as well as other gases
- A solution is a homogeneous mixture made up of:
  - The substance being dissolved, the **solute**
  - A medium of dissolution, the **solvent**
- Solutions exist in three physical states:
  - Solids, e.g alloys
  - Liquids, e.g sodium chloride in water
    - Liquid solutions where the solute is an ionic compound and the solvent is water are known as **aqueous solutions**
  - Gaseous, e.g air
- A **colloid** is similar to a solution in that it appears to be homogeneous like a solution
  - However, it consists of comparatively large particles of one substance dispersed throughout another substance or solution

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Molarity

- A solution may be described **qualitatively or quantitatively**
  - Qualitatively, the terms **dilute** and **concentrated** are used to describe the amount of solute in a given quantity of the solvent
    - A solution with a relatively small concentration of solute is said to be dilute
    - A solution with a large concentration of solute is said to be concentrated
  - Quantitatively, a solution is described in terms of its **concentration**

Molarity

- Molarity (M) expresses the concentration of a solution as the number of moles of solute in a litre of solution:

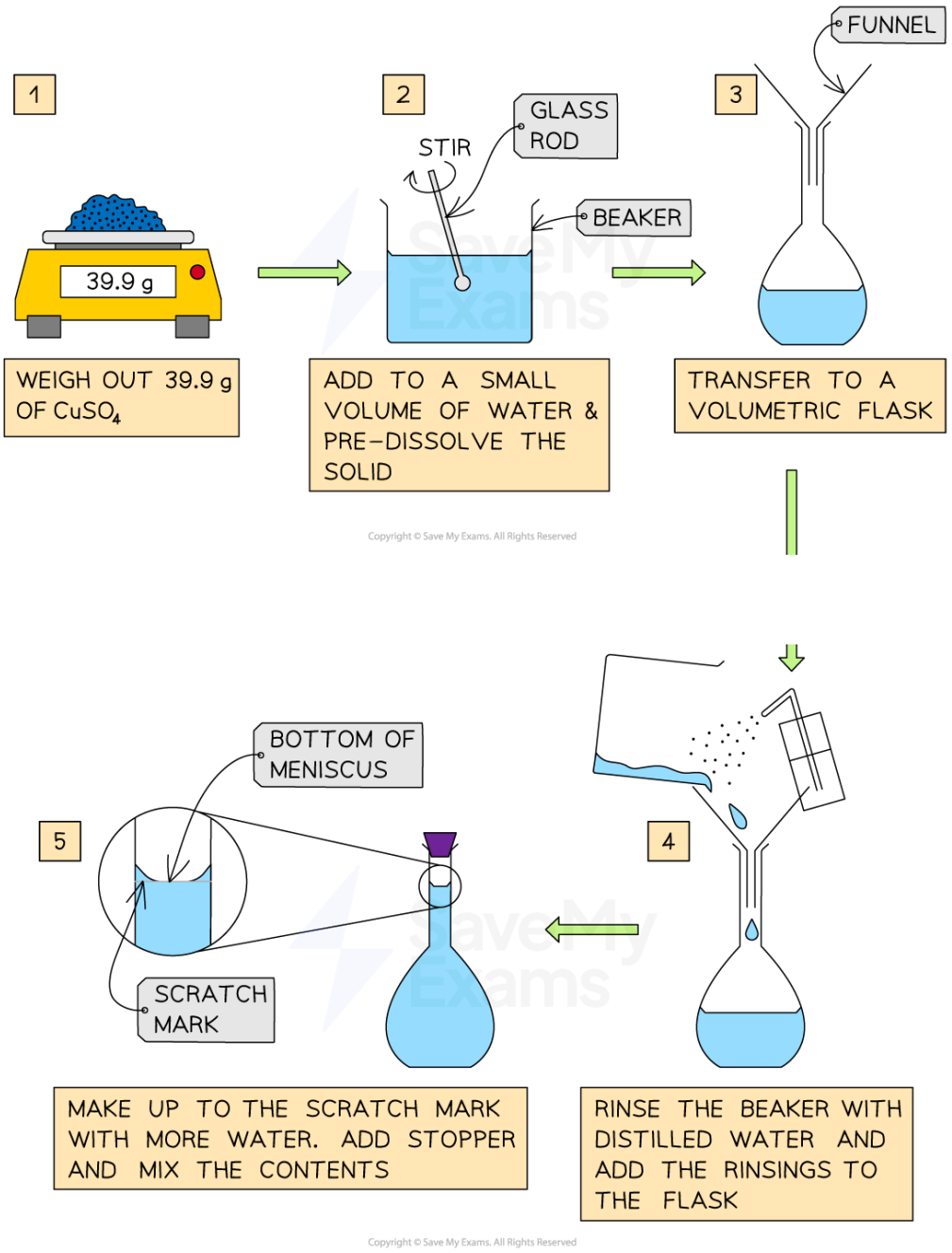
$$\text{Molarity (M)} = \text{Moles of Solute (n)} / \text{Volume of Solution (L)}$$

$$M = n / V$$

- A **1.00 molar solution** (1.00 M) contains 1.00 mol of solute in every 1.0 L of solution
- A solution can be prepared to a specified molarity by weighing out the calculated mass of solute and dissolving it in enough solvent to form the desired volume of solution
- For example, to prepare 250.0 mL of a 1.00 M solution of $CuSO_4$:
  - Determine the number of moles of the solute ($CuSO_4$) required
    - In this case, this is given as $n = Molarity \times Volume$
    - $n = 1.00 \times 0.250 = 0.250$ mol
  - Determine and weigh the corresponding mass of $CuSO_4$ required
    - Using the expression: $Mass = n \times M_r$
    - The molar mass (M_r) of $CuSO_4$ is calculated as:
    - $M_r = (M_r)_{Cu} + (M_r)_S + 4(M_r)_O$
    - $M_r = 63.6 + 32.0 + 4\,(16) = 159.6$ g/mol
    - Mass = 0.250 mol × 159.6 g/mol = 39.9 g
    - Transfer 39.9 g of $CuSO_4$ into a 250 mL flask and add some quantity of water to dissolve the solute
    - Add more water until the solution reaches the calibrated mark of the flask

Preparing a Solution

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How to prepare a 250mL of 1.00 M solution of $CuSO_4$

Dilution

- Alternatively, you can start with a more concentrated solution, called the **stock solution,** and dilute it with water to give a solution of the desired molarity
  - The calculations are straightforward if you keep a simple point in mind: Adding solvent **cannot** change the number of moles of solute
    - That is:

n_{solute} *(stock solution)* = n_{solute} *(dilute solution)*

- In both solutions, *n* can be found by multiplying the molarity, M, by the volume in litres, V
- Hence, a dilution expression can be obtained:

$$M_s \times V_s = M_d \times V_d$$

- Where the subscripts *s* and *d* stand for stock and dilute solutions, respectively



Your notes

Worked Example

Calculate the molarity of a solution made by dissolving 5.00 g of glucose ($C_6H_{12}O_6$) in sufficient water to form exactly 100 mL of solution. (M_r glucose = 180.2 g/mol)

Answer:

Analyse: We are provided with the mass of glucose and the volume of solution and asked to determine the molarity of the solution

Plan: To determine the molarity of the solution the following steps are required:

Step 1: Convert the mass of glucose to moles and the volume of the solution to litres

- *n* = Mass/Molar Mass
- *n* = 5.00/180.2
- *n* = 0.0277 moles
- Volume = 100/1000
- Volume = 0.1L

Step 2: Substitute the values of the moles of glucose and the volume of the solution

- Molarity (M) = number of moles/volume of solution
- Molarity (M) = 0.0277/0.1
- Molarity (M) = 0.277 M



Worked Example

1. How many grams of Na_2SO_4 are there in 15 mL of 0.50 M Na_2SO_4?
2. How many millilitres of 0.50 M Na_2SO_4 solution are needed to provide 0.038 mol of this salt?

Analyse:

- In part **a**, we are given the volume of the solution and the concentration (Molarity)
  - We are then asked to determine the mass of the solute(Na_2SO_4) required
- In part **b**, we are given the number of moles of solute and the molarity of the solution
  - We are then asked to determine the volume of solution in millilitres

Plan:

For part **a**:

- Use the molarity expression to determine the number of moles of the solute (n)
- Convert this number of moles to mass

For part **b**:

- Use the molarity expression to determine the volume of solution in litres
- Convert to millilitres using the appropriate conversion factor

Answer:

- Part **a**:
  - **Step 1:** Convert volume in mL to L:
    - Volume of solution = 15/1000 = 0.015L
  - **Step 2:** Rearrange the concentration expression in terms of the number of Na_2SO_4 (*n*):
    - $n = M \times V$
    - $n = 0.50 \times 0.015$
    - $n = 0.0075$ moles
  - **Step 3:** Using the mole-molar mass expression, convert the moles of solute to mass:
    - $M_{Na2SO4} = n \times M_r$
    - $M_{Na2SO4} = 0.0075 \times 142.04$
    - $M_{Na2SO4} = 1.1$g
- Part **b**:
  - **Step 1:** Rearrange the concentration expression in terms of the volume of solution:
    - $V = n/M$
    - $V = 0.038/0.50$
    - $V = 0.076$ L
  - **Step 2:** Convert volume in L to mL:
    - $V = 0.076 \times 1000$
    - $V = 76$ mL

Worked Example

How many millilitres of 5.0 M $Na_2Cr_2O_7$ solution must be diluted to prepare 250 mL of 0.10 M solution? How much water needs to be added to this concentrated solution?

Analyse:

- We are given the volume and concentration of the diluted solution and asked to determine the volume of the concentrated solution required to prepare
- We are then asked the volume of water required to make the dilute solution

Plan:

- Using the dilution formula, we can calculate the volume of the concentrated solution required

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- Then subtract that volume from the volume of the dilute solution to obtain the volume of water required

Answer:

- **Step 1:** Using $M_1V_1 = M_2V_2$
  - Rearrange the equation to $V_1 = (M_2V_2)/ M_1$
    - $V_1 = ?$
    - $M_1 = 5.0$ M
    - $V_2 = 250$ mL
    - $M_2 = 0.10$ M
  - So, $V_1 = (0.10 \times 250)/5.0$
  - $V_1 = 5.0$ mL
- **Step 2:** The volume of water added will be:
  - $V_{H2O} = 250 - 5.0$
  - $V_{H2O} = 245$ mL



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Your notes

Particulate Representations of Solutions

- A **solution** is a **homogeneous** mixture of two substances
- The components of a solution are a **solute** and a **solvent**
  - The **solvent** is the component that is in the highest amount
    - **Liquids** usually act as solvents in solutions
  - The **solute** is the component that is in the least amount
- E.g. A mixture of salt water is the best example for a solution. Salt is the solute and water acts as a solvent
- Any solution in which water acts as a solvent is called an **aqueous solution**
- **Concentration** makes reference to the amount of particles of solvent into an specific volume of solute
  - The greater the amount of solute particles in a fixed volume, the greater the concentration
- A **concentrated solution** has a large amount of solute particles
- A **diluted solution** has a relatively small amount of solute



Examiner Tips and Tricks

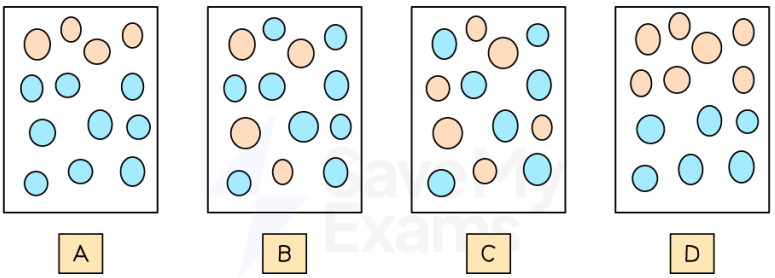
Solutions are homogeneous mixtures, therefore the molecules of solute and solvent must be evenly spread in the container without forming regions. If individual regions of solute and solvent are formed, it is not a solution



Worked Example

Which of the following samples of a salt + water solution has the highest concentration?

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Answer:

The correct answer is **C** because:

- The particles of solute and solvent are evenly spread in the container
- It has a largest amount of solute particles (salt particles) compared to **B** which is also a solution
- **A** and **D** are not correct. The particles are forming individual regions of solute and solvent. Therefore, they are **heterogenous mixtures** which are not solutions

Your notes

Paper Chromatography

Principles of Paper Chromatography

- A solution is a **mixture** that cannot be separated by filtration, therefore an alternative separation technique must be used
- **Paper chromatography** is used to separate a mixture of liquids that have different intermolecular interactions in a given solvent (e.g. separate the coloured inks that were used to make black ink)
- The substances can be identified by the distance they travel across a sheet of **chromatography paper**
- The stronger the interaction with the solvent, the farther the distance the component will travel up the chromatography paper
- The solvent will travel up the paper because the liquid molecules will have adhesive and cohesive forces with the pores of the chromatography paper
  - Since the solvent moves up the paper, the solvent is called the **mobile phase**
  - Since the chromatography paper stays in a fixed position, it is called the **stationary phase**
- The different components of the mixture will be separated because of the different intermolecular interactions with the mobile and the stationary phases

Conducting a Paper Chromatography analysis

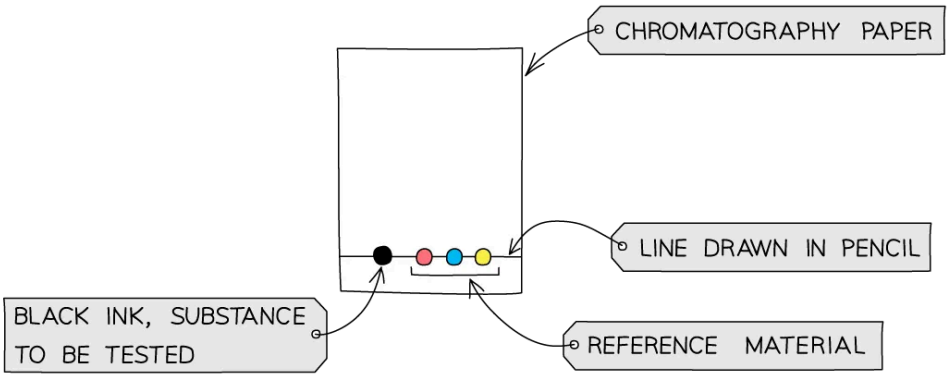
1. Draw a **pencil line** at the bottom of the chromatography paper. A sample of the mixture is placed as a spot on the top of the line. Additional spots of known reference compounds are placed to identify the components of the mixture
2. Place the paper into the solvent container, making sure that the line sits **above** the level of the solvent. This will prevent the samples from washing into the solvent container
3. Analyze the chromatography paper by comparing the sample with the reference compounds

Diagram to Show the Method of Chromatography

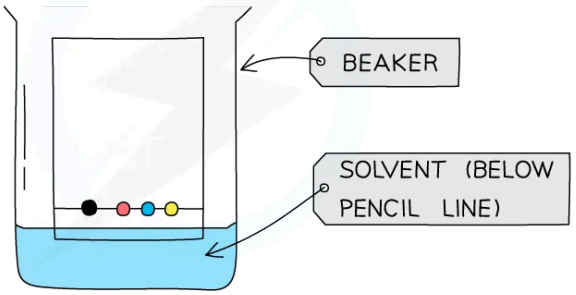
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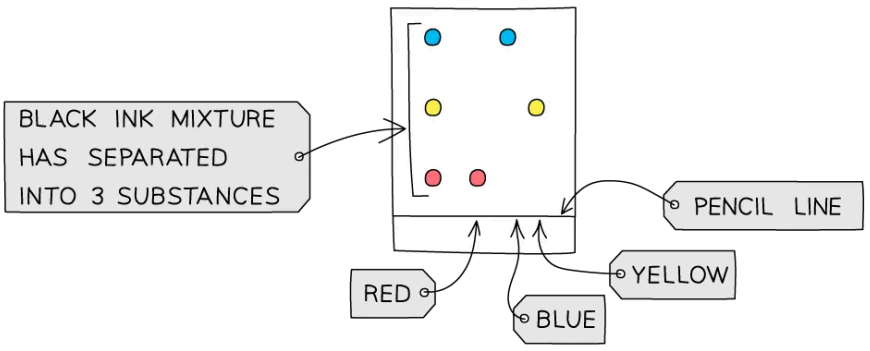
1 SET UP CHROMATOGRAPHY PAPER AS SHOWN



2 LOWER PAPER INTO A BEAKER WITH APPROPRIATE SOLVENT. WAIT FOR SOLVENT TO TRAVEL UP THE PAPER.



3 ANALYSE CHROMATAGRAM



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Analysis of the composition of ink that can be used using paper chromatography. Blue ink travels the most distance showing the strongest interaction with the solvent and the weakest interaction with the chromatography paper

Your notes

- The component that travels the **greatest distance** along the paper has the **strongest intermolecular interaction with the mobile phase** and the **weakest interaction with the stationary phase**
- The component that travels the **shortest distance** along the paper has the **weakest intermolecular interaction with the mobile phase** and the **strongest interaction with the stationary phase**

Thin Layer Chromatography

Principles of Thin Layer Chromatography

- Thin Layer Chromatography (TLC) is a separation technique used to analyze the components of a mixture by exploiting their different affinities towards a material
- In the same way as paper chromatography, there are two phases involved in TLC: an stationary phase and a mobile phase

Commonly used stationary phases

- The most common material is a thin metal sheet coated in silica (SiO_2) or alumina (Al_2O_3)
- The different components from the mixture will be separated because of the different intermolecular interactions with the coating
- Each component will travel a distance that depends on the interaction with the stationary phase

Commonly used mobile phases

- It could be polar or nonpolar liquids
  - Polar solvents (water or alcohol)
  - Nonpolar solvents (liquid alkanes)
- The mobile phase will flow over the stationary phase by adhesive and cohesive forces with the mobile phase

Conducting a TLC analysis

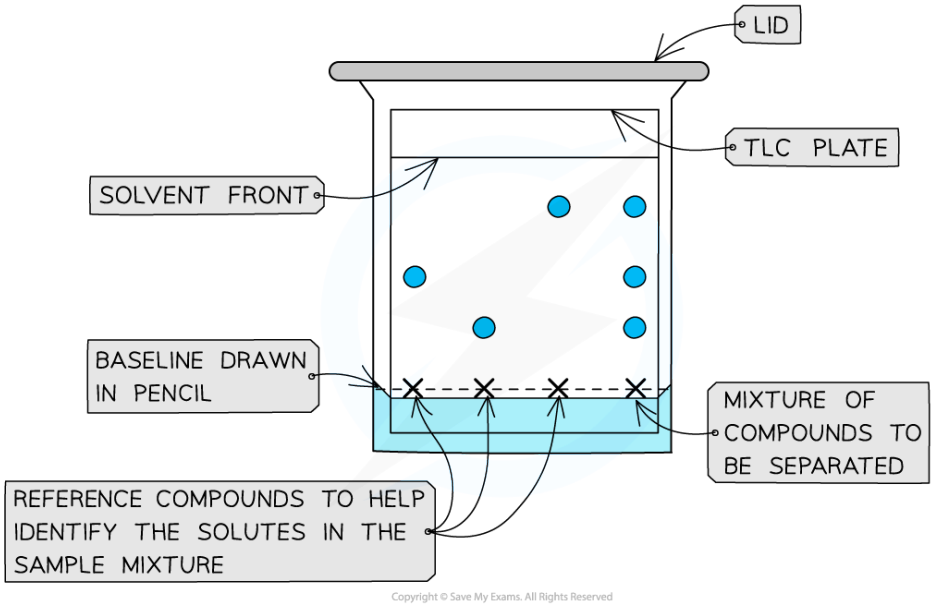
1. Draw a **pencil line** at the bottom of a TLC plate
2. A sample of the mixture is placed as a spot on the top of the line. Additional spots of known reference compounds are placed to identify the components of the mixture
3. Place the TLC plate into the solvent container and place a lid to cover the beaker, making sure that the line sits **above** the level of the solvent. This will prevent the samples from washing into the solvent container
4. The solvent will begin to travel up the plate

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5. When the solvent almost reaches the top, draw another pencil line where the solvent has reached. This indicates the **solvent font**
6. Analyze the TLC plate and identify the components of the mixture by using the reference compounds

Your notes

Method for TLC Analysis



A dot of the sample is placed on the baseline and allowed to separate as the mobile phase flows through the stationary phase; The reference compound/s will also move with the solvent

- The component that travels the **most distance** has the **strongest intermolecular interaction with the mobile phase** and the **weakest interaction with the stationary phase**
- The component that travels the **less distance** has the **weakest intermolecular interaction with the mobile phase** and the **strongest interaction with the stationary phase**

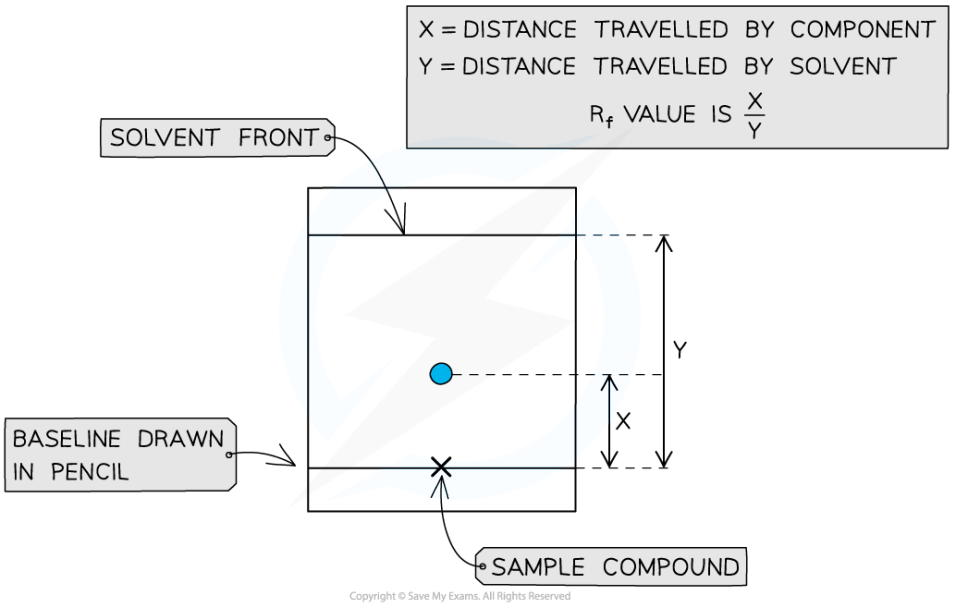
Rf values

- The TLC plate can be analyzed to calculate the Rf values for each compound

$$R_f = \frac{\text{Distance traveled by the component}}{\text{Distance traveled by the solvent}}$$

- These values gives important information about the identity of the components and their polarities
- Since the **stationary phase is polar**, the higher the R_f values the more polar the component
- The R_f value and the difference between the separated substances will also depend on their solubility in the solvent that acts as the **mobile phase**

Calculating the R_f value



R_f values can be calculated by taking 2 measurements from the TLC plate

Column Chromatography

Principles of Column Chromatography

- Column chromatography is the most sophisticated chromatography technique
- In the same way as paper chromatography and TLC, it separates a mixture in its components by exploiting their different affinities towards a stationary phase and a mobile phase
- This technique takes more time, but it allows to separate

Commonly used stationary phases

- In Column chromatography, silica (SiO_2) or alumina (Al_2O_3) are packed in the column
- The different components from the mixture will be separated because of the different intermolecular interactions with the polar packaging
- Each component will have different movement rates that depends on the interaction with the stationary phase

Commonly used mobile phases

- It could be polar or nonpolar liquids, and they need to be compatible with the packing of the
- The mobile phase will flow down the stationary phase

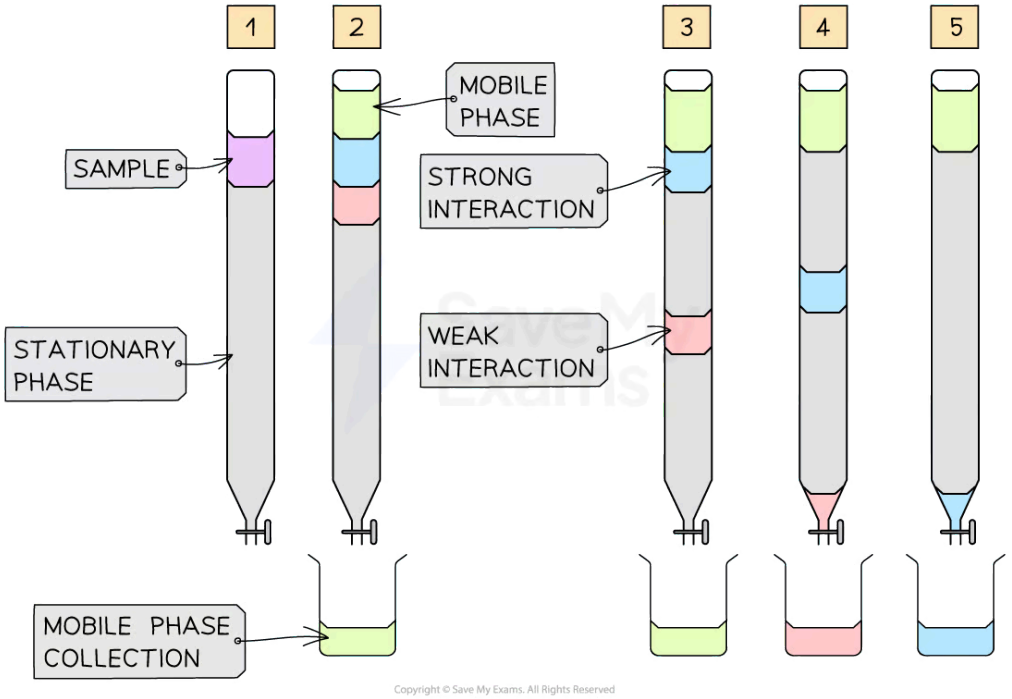
Conducting a Column Chromatography analysis

Your notes

1. Pack the column with a stationary phase and ensure an even packing to facilitate uniform flow. Dissolve the sample in a solvent that is compatible with the stationary phase. Apply the sample onto the top of the column using a pipette, ensuring a concentrated and narrow band
2. Choose an appropriate solvent (mobile phase). Pour the solvent mixture into the column slowly to allow the sample to start moving down the column
3. As the sample moves down the column, different components will separate based on the interactions for the stationary phase
4. Collect the different mixture components in tubes or vials
5. Label and record all the components of the sample for later analysis

Your notes

Column Chromatography



Steps for performing a Column chromatography analysis

Your notes

Separation by Distillation

Principles of Distillation

- A liquid solution is mixture of **two miscible liquids** that cannot be separated through filtration, therefore, an alternative separation technique must be used
- **Distillation** is a technique that exploits the differences in boiling points between the components of a liquid solution
- The boiling point of the liquids is influenced by the intermolecular forces within the molecules of the substance
  - The **stronger the intermolecular forces**, the **higher the boiling point** of the liquid

Conducting a separation by distillation

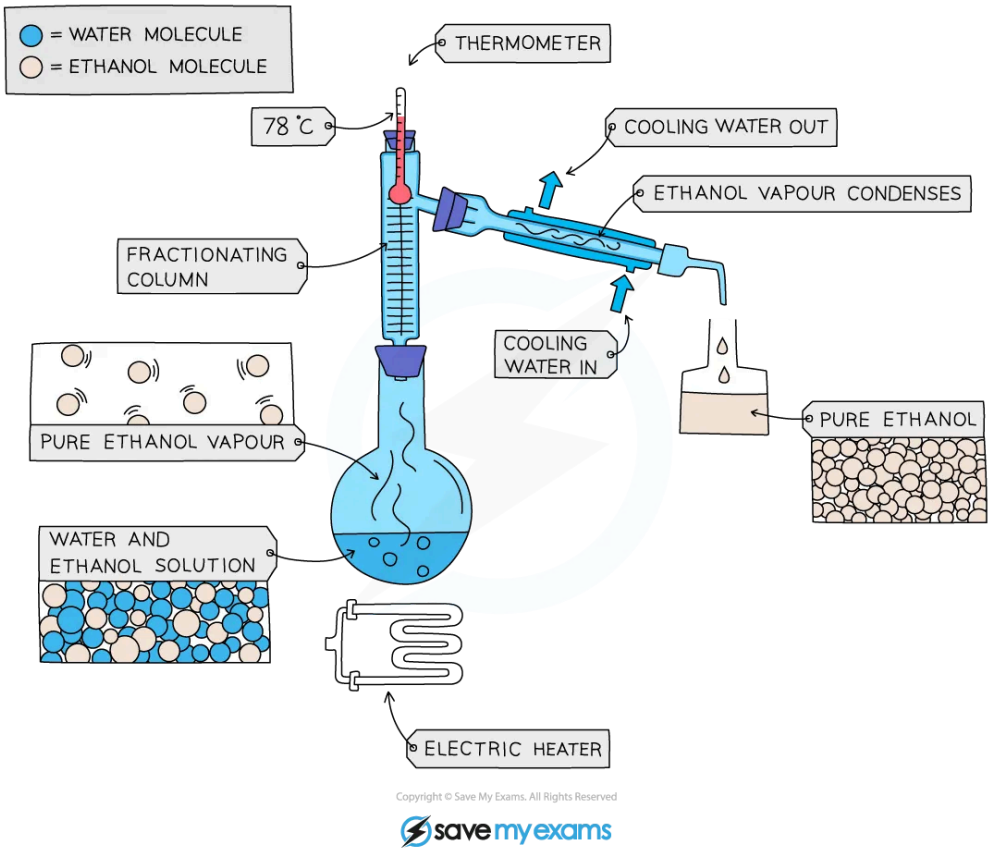
- A distillation of a liquid solution of water-ethanol can be carried in simple steps:

1. Place the mixture in a round-bottom flask, and connect it to a fractionating column followed by a condenser
2. Apply heat to the flask to increase the vapor pressure of the components in the mixture. Using an electric heater is mandatory when flammable liquids are present in the mixture
3. When the vapor pressure of one of the components of the mixture is equal to the atmospheric pressure, it will boil. This is the **most volatile** compound and it vaporizes first because of its **weak intermolecular forces**
4. The vapor passes through the condenser, where it cools down, turning itself into a liquid again
5. As the vapor condenses, the component with the stronger intermolecular forces, remains in the round bottomed flask
6. Finally, the cooled liquid is collected in a separate container

- For water and ethanol, ethanol has a boiling point of 78 °C and water of 100 °C. The mixture is heated until it reaches 78 °C, at which point the ethanol boils and distills out of the mixture and condenses into the beaker
- The temperature should not reach 100 °C, to prevent water molecules to evaporate
- The fractional column is not necessary if there is a big difference between the boiling points of the components from the mixture

Method of Separation by Distillation

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Separation of a mixture of ethanol and water is best achieved by fractional distillation when the two components are close together in boiling point or there are multiple components

Your notes

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Solubility of Ionic & Molecular Compounds

- The solubility of two substances depends on their **intermolecular interactions**
- Ions and molecules can exhibit can exhibit the following interactions:

Relative strength comparison between the intermolecular interactions

Interaction	Relative strength
Ion-dipole	Strongest
Hydrogen bonding	
Dipole-dipole	
London dispersion forces	Weakest

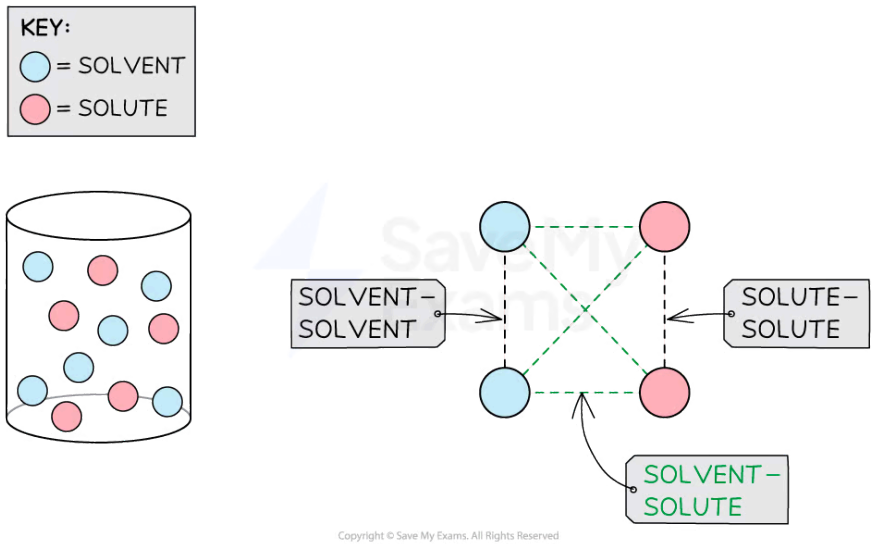
- A solution is made from two components a **solvent** and a **solute**
- In a solution system, intermolecular interactions exists in three possible scenarios:
  - Solvent-solute, i.e solvent particles with solute particles
  - Solvent-solvent, i.e. solvent particles with themselves
  - Solute-solute, i.e.solute particles with themselves

When does a solution form?

- A solution is formed when solvent-solvent interactions and the solute-solute interactions are the same
  - When this happens, the substances are said to be **miscible**
  - E.g. Ethanol and water present hydrogen bonding interactions. Therefore, they are miscible in one another
- When the solvent-solvent interaction and the solute-solute interaction are not equal, the strength of the solvent-solute interaction must be considered

The three interactions between the particles of solute and solvent

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Solvent-solvent, solvent-solute and solute-solute interactions must be considered for solubility

- A solution is formed when the solvent-solute interactions are stronger than the solvent-solvent interactions
  - E.g. Salt and water present ion-dipole interactions. Water molecules present hydrogen bonding between themselves

ion-dipole > hydrogen bonding

- Therefore, they **are miscible** in one another
- A solution will not be formed when the solvent-solute interactions are weaker than the solvent-solvent interactions
  - E.g. Hexane and water present London Dispersion Forces. Water molecules present hydrogen bonding themselves

London-Dispersion Forces < hydrogen bonding

- Therefore, they are **not miscible** in one another
- The rules of solution formation are summarized in the table below

Rules of Solution Formation by comparing the intermolecular interactions strengths

Comparison of the strength between intermolecular interactions	Solution formed?
Solvent-solute = Solvent-solvent	Yes
Solvent-solute > Solvent-solvent	Yes

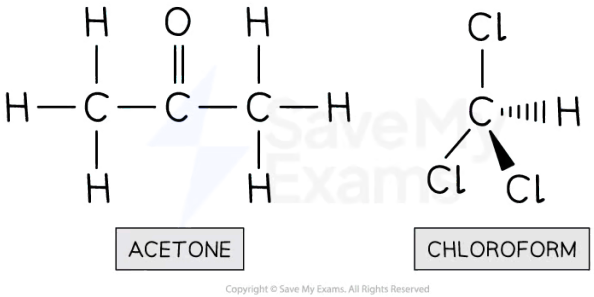
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Solvent-solute < Solvent-solvent	May or may not form

Your notes

Worked Example

Predict if a solution will be formed when chloroform is mixed with an acetone solvent.

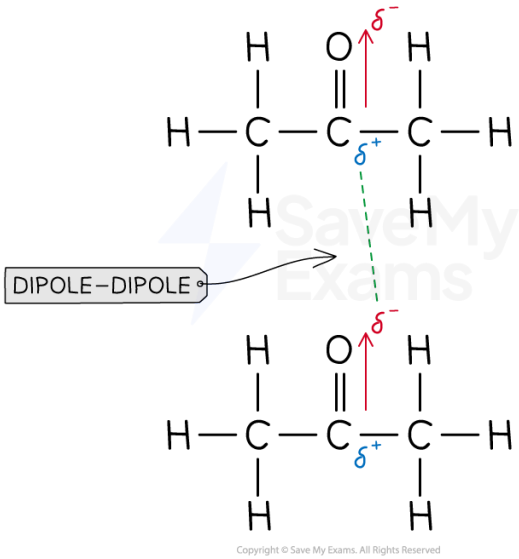


Answer:

Step 1 (Optional): Draw the molecular structure of the solvent molecules and the solute molecules. This step is not applicable to ions since they exist as lattice structures

Step 2: Identify the solvent-solvent intermolecular interactions

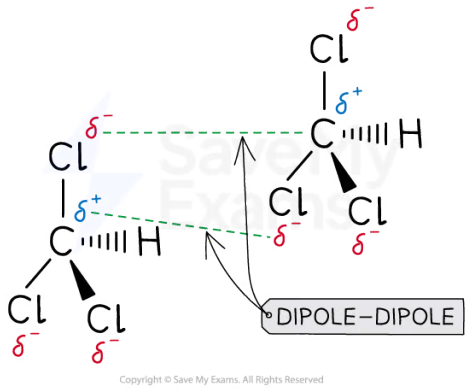
- Acetone is our solvent. Since oxygen is more electronegative than carbon, a dipole moment will be formed



- Therefore, the solvent-solvent intermolecular interaction is **dipole-dipole**

Step 3: Identify the solute-solute intermolecular interactions

- Chloroform is our solute. Since it is a non-symmetrical molecule and chlorine is more electronegative than carbon, dipole moments will be formed



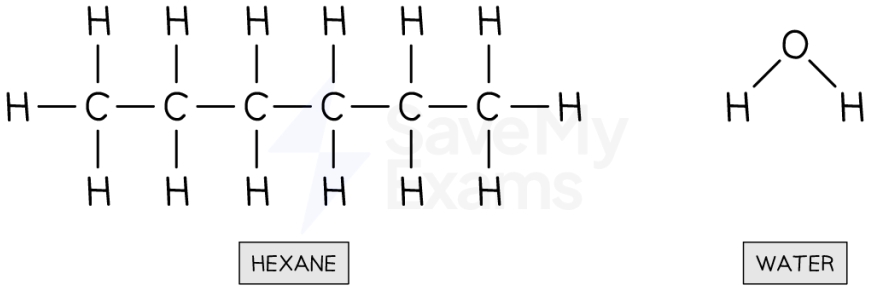
- Therefore, the solute-solute intermolecular interaction is **dipole-dipole**

Step 4: If the solute-solute interaction and the solvent-solvent interactions are the same, they are miscible. Otherwise, the table should be used and the solvent-solute interactions must be considered

- Since acetone presents dipole-dipole interactions and chloroform presents dipole-dipole interactions, both solute and solvent will be miscible in one another

Worked Example

Predict if a solution will be formed when hexane is mixed with water as solvent.



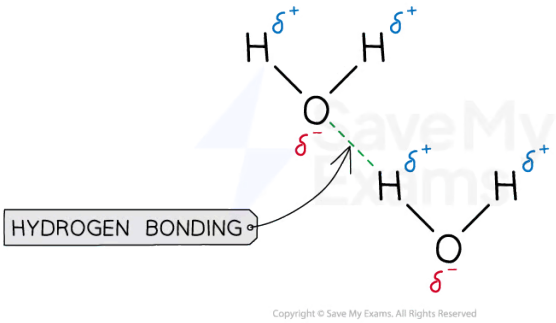
Answer:

Step 1 (Optional): Draw the molecular structure of the solvent molecules and the solute molecules. This step is not applicable to ions since they exist as lattice structures

Step 2: Identify the solvent-solvent intermolecular interactions

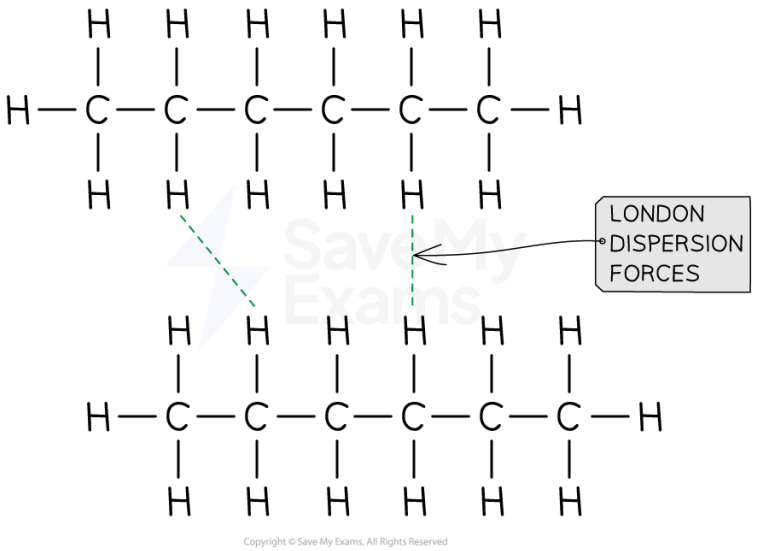
- Water is our solvent. Since hydrogen is bonded to a highly electronegative atom such as oxygen, a strong dipole moment will be formed
- Therefore, the solvent-solvent intermolecular interaction is **hydrogen bonding**

Your notes



Step 3: Identify the solute-solute intermolecular interactions

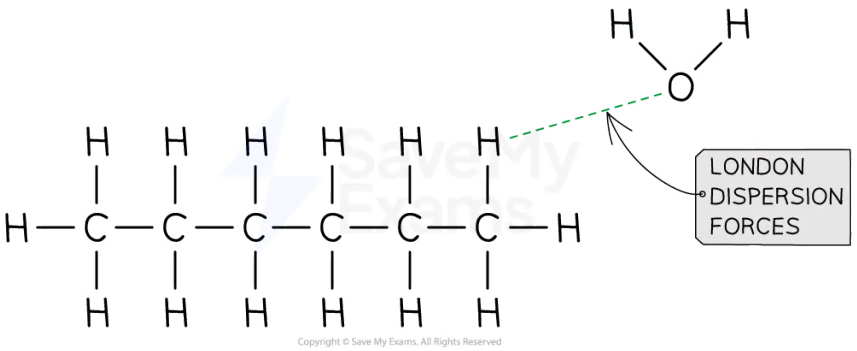
- Hexane is our solute. Since hexane is a symmetrical molecule, all its dipole moments are canceled out



- Therefore, the solute-solute intermolecular interaction are **London dispersion forces**

Step 4: If the solute-solute interaction and the solvent-solvent interactions are the same, they are miscible. Otherwise, the table should be used and the solvent-solute interactions must be considered

- The solvent-solvent interaction is not the same as solute-solute interaction
- The solvent-solute interaction must be considered



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- As shown in the image below, hexane and water present London dispersion forces interactions because oxygen induces momentaneous dipoles in hexane
- Using the rules of solution, if solvent-solute < solvent-solvent, a solution will not be formed
- Since London-dispersion forces < hydrogen bonding in terms of relative strength, **hexane and water are not miscible,** and a **solution will not be formed**

Your notes